A Hybrid Automated Essay Scoring Framework Integrating DeBERTa Embeddings and Handcrafted Linguistic Features

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Abstract - *Automated Essay Scoring (AES) has advanced significantly with transformer-based language models, yet a tension remains between predictive accuracy and pedagogical interpretability. This paper presents a multi-granularity, linguistically enriched AES framework that combines frozen transformer-based semantic representations with interpretable handcrafted linguistic features and a lexical (TF-IDF) signal. Essays are modeled at token, sentence, paragraph, and chunk levels, and the representations are fused by a BiLSTM-attention network. To avoid information leakage, the network and the lexical model are trained out-of-fold, and their predictions are stacked with the linguistic features in gradient-boosted regressors (LightGBM and XGBoost), followed by cut-points optimized for Quadratic Weighted Kappa (QWK). Evaluated on the Learning Agency Lab Automated Essay Scoring 2.0 dataset (17,307 essays, scores 1 to 6), the model achieves a QWK of 0.8206 on a held-out test set of 1,731 essays (95% bootstrap CI 0.8030 to 0.8367). Ablations show the contribution of each component. Beyond scoring, the system generates personalized feedback from interpretable linguistic indicators.*

1. Introduction

1.1 Background and Problem Statement

Essay writing is widely used to assess critical thinking and linguistic competence, but manual grading is time-consuming and subjective, with scores varying across examiners. As online assessments expand, scalable and consistent scoring has become essential. Automated Essay Scoring (AES) addresses this by predicting human-assigned scores. Early systems relied on

interpretable handcrafted features such as word count, lexical richness, and discourse structure, but struggled to capture semantic context. Transformer-based models like BERT and DeBERTa improved semantic modeling, yet at the cost of interpretability. Moreover, most AES systems operate at a single textual granularity or use only one information source, overlooking how local writing quality relates to broader discourse structure, and they offer little actionable feedback. This motivates a framework that integrates linguistic and semantic information across multiple textual levels for accurate scoring with interpretable feedback.

1.2 Related Works

AES research has moved from handcrafted linguistic and statistical features, such as the coherence, cohesion, and lexical features of Alawadh et al. [1], to deep learning models like CNN-BiGRU [6] and Sentence-BERT with BiLSTM-attention [7]. Transformer-based methods further improved contextual representation [2], while trait-based and feedback-oriented systems addressed interpretability and pedagogical value [3, 4]. Concerns about fairness, transparency, and reliability have also been raised [5, 8]. Still, most systems prioritize accuracy and rely on either linguistic features or semantic representations at a single granularity, which limits interpretability and coverage of essay structure. This gap motivates our framework, which combines interpretable linguistic indicators with multi-granularity semantic representations for scoring and feedback.

1.3 Contributions

This study proposes a multi-granularity, linguistically enriched AES framework that integrates frozen DeBERTa and MiniLM representations at the token, sentence, paragraph, and chunk levels with seventeen interpretable linguistic features. Three design choices distinguish it. First, the BiLSTM fusion network is trained out-of-fold so that downstream models never see predictions made on essays the network was trained on. Second, a lexical TF-IDF branch and the

fusion network are stacked with the linguistic features in LightGBM and XGBoost regressors. Third, final scores are obtained with cut-points that maximize QWK on out-of-fold training predictions. The system also generates targeted feedback from interpretable indicators. It is evaluated on the Learning Agency Lab Automated Essay Scoring 2.0 dataset using QWK, with ablations that isolate each component.

2. Dataset and Methods

2.1 Dataset

Experiments use the public Learning Agency Lab Automated Essay Scoring 2.0 training corpus (Kaggle) [9], which contains 17,307 student-written argumentative essays. Each essay has a human-assigned holistic score on an integer 1 to 6 scale (score counts: 1: 1,252; 2: 4,723; 3: 6,280; 4: 3,926; 5: 970; 6: 156). No essay is duplicated. The data were split once, stratified by score, into 13,845 training, 1,731 validation, and 1,731 test essays (80:10:10). The validation split is used only for early stopping of the neural network, and the test split is used only for the final evaluation.

2.2 Text Preprocessing and Feature Extraction

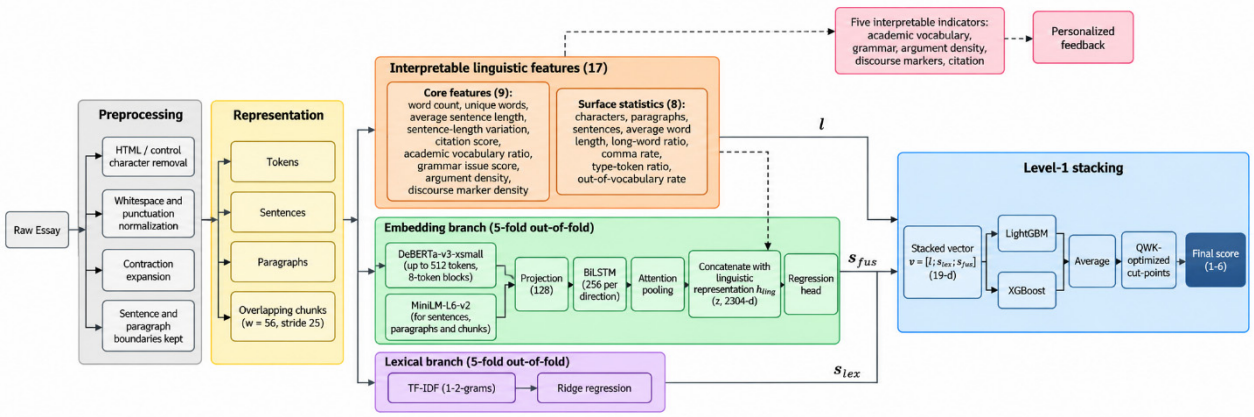


Figure 1: Architecture of the proposed multi-granularity hybrid AES framework

Each raw essay x_i was normalized to reduce noise while preserving structure: HTML tags and control characters were removed, whitespace and punctuation spacing were standardized, contractions were expanded, and paragraph and sentence boundaries were retained.

Essays were represented at four granularities: tokens, sentences, paragraphs, and overlapping chunks. Sentences were split on sentence-ending punctuation followed by whitespace, or on newlines, and paragraphs on newline-separated text. Chunks were produced by sliding a window of size w and stride s over the token sequence:

$$C_j = [t_{s(j-1)}, \ldots, t_{s(j-1)+w-1}] \quad \ldots(1)$$

where t_k is the *k-th* token (tokens are indexed from 0). The overlap lets chunks retain context.

In parallel, a seventeen-dimensional interpretable feature vector li $\in \mathbb{R}^{17}$ was extracted for each essay:

$$\mathbf{l}_i = [\mathrm{WC}, \mathrm{UWC}, \mathrm{ASL}, \mathrm{SLV}, \mathrm{CS}, \mathrm{AVR}, \mathrm{GIS}, \mathrm{AD}, \mathrm{DMD}, \mathrm{NC}, \mathrm{NP}, \mathrm{NS}, \mathrm{AWL}, \mathrm{LWR}, \mathrm{CR}, \mathrm{TTR}, \mathrm{OOV}] \ldots\ldots (2)$$

The first nine are word count (WC), unique word count (UWC), average sentence length (ASL), sentence-length variation (SLV), citation score (CS), academic vocabulary ratio (AVR), grammar issue score (GIS), argument density (AD), and discourse marker density (DMD). The remaining eight are surface statistics: character count (NC), paragraph count (NP), sentence count (NS), average word length (AWL), long-word ratio (LWR; words of seven or more letters), comma rate (CR), type-token ratio (TTR), and out-of-vocabulary rate (OOV; the share of words not seen at least five times in the training essays, a proxy for misspelling). All statistics, including the vocabulary and the standardization parameters, were computed on the training split only. The vector is used for scoring and, through five of its indicators, for feedback generation.

2.3 Multi-Granularity Representation and Fusion

The four representations from Section 2.2 were embedded independently with frozen encoders. Token-level embeddings came from microsoft/deberta-v3-xsmall [10], which reads up to 512 tokens; hidden states were averaged in non-overlapping blocks of eight tokens, giving at most 64 vectors per essay. Sentence-, paragraph-, and chunk-level embeddings came from all-MiniLM-L6-v2 [11, 12]. Let $E_w^{(i)}, E_s^{(i)}, E_p^{(i)}, and\ E_c^{(i)}$ denote the resulting sequences.

Because the encoders output different dimensions, each sequence passed through a learned linear projection to 128 dimensions and then a BiLSTM with hidden size 256 per direction, giving 512-dimensional contextual states. For $\ell \in \{w, s, p, c\}$:

$$H_\ell = \mathrm{BiLSTM}\big(\mathrm{Proj}_\ell(E_\ell)\big) \quad \text{...(3)}$$

Attention pooling then emphasized the most informative elements of each sequence:

$$\alpha_{\ell,t} = \frac{\exp(v_\ell^T \tanh(W_\ell h_{\ell,t}))}{\sum_{k=1}^{T_\ell} \exp(v_\ell^T \tanh(W_\ell h_{\ell,k}))} \quad \text{…(4)}$$

$$e_\ell = \sum_{t=1}^{T_\ell} \alpha_{\ell,t}\, h_{\ell,t}, \quad \ell \in \{w, s, p\} \quad \text{…(5a)}$$

$$\mathbf{h}_{\mathrm{attn}} = \sum_{t=1}^{T_c} \alpha_{c,t}\ \mathbf{h}_{c,t} \quad \text{…(5b)}$$

$$\bar{\mathbf{e}}_c = \frac{1}{T_c} \sum_{t=1}^{T_c} \mathrm{Proj}_c\left(\mathbf{E}_{c,t}\right) \quad \text{…(5c)}$$

Padding received zero attention weight. Attention pooling of the word, sentence, and paragraph states gives e_w, e_s, and e_p; pooling the chunk states gives h_{attn}. In addition, the chunk branch keeps $\bar{e}_c$, the mean of the projected (128-dimensional) chunk embeddings. The linguistic vector l_i was passed through a linear layer with ReLU to give a learned linguistic representation:

$$h_{\mathrm{ling}} = \mathrm{ReLU}\big(W_{\mathrm{ling}} l_i + b_{\mathrm{ling}}\big), \qquad h_{\mathrm{ling}} \in \mathbb{R}^{128} \quad \text{…(6)}$$

All representations were concatenated into a unified essay representation:

$$z_i = \left[h_{\text{ling}} \parallel h_{\text{attn}} \parallel e_w \parallel e_s \parallel e_p \parallel \bar{e}_c\right], \qquad z_i \in \mathbb{R}^{2304} \qquad \ldots(7)$$

A small feed-forward head (one 128-unit hidden layer with dropout 0.2) maps z_i to the fusion prediction $\hat{y}_i^{\text{fus}}$, and an auxiliary head maps h_{ling} to $\hat{y}_i^{\text{ling}}$.

2.4 Training and Stacked Scoring

The fusion network was trained with the loss

$$\mathcal{L}_{\text{main}} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i^{\text{fus}}\right)^2 \qquad \ldots(8)$$

$$\mathcal{L}_{\text{ling}} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i^{\text{ling}}\right)^2 \qquad \ldots(9)$$

$$\mathcal{L}_{\text{fusion}} = \mathcal{L}_{\text{main}} + \alpha\mathcal{L}_{\text{ling}} \qquad \alpha = 0.15 \qquad \ldots(10)$$

using Adam with weight decay and gradient clipping, with early stopping on validation RMSE.

To prevent leakage, scoring is stacked. Within the training split, the fusion network was trained in five folds; each essay's prediction s_i^{fus} (the out-of-fold value of $\hat{y}_i^{\text{fus}}$) comes from the model that did not see it. A lexical branch was trained the same way: TF-IDF features (word 1- and 2-grams, minimum document frequency 3, sublinear term frequency, at most 120,000 features, vocabulary fitted on training essays only) were fed to a ridge regressor, giving s_i^{lex}. Validation and test essays received the average of the fold models' predictions.

Level-1 models then used the stacked vector

$$\mathbf{v}_i = \left[\mathbf{l}_i \parallel s_i^{\text{lex}} \parallel s_i^{\text{fus}}\right] \in \mathbb{R}^{19} \qquad \ldots(11)$$

and a LightGBM [13] and an XGBoost [14] regressor were trained on it, with the number of boosting rounds chosen by inner five-fold cross-validation. Their predictions were averaged:

$$\hat{y}_i = \frac{1}{2}\, g_{\text{LGBM}}(\mathbf{v}_i) + \frac{1}{2}\, g_{\text{XGB}}(\mathbf{v}_i) \qquad \ldots. (12)$$

Finally, the continuous score was mapped to the 1 to 6 scale with cut-points $c_1 < c_2 < \ldots < c_5$ chosen by coordinate search to maximize QWK on the out-of-fold training predictions:

$$\hat{y}_i^{\text{final}} = 1 + \sum_{k=1}^{5} I\,(\hat{y}_i \geq c_k), \quad c_1 < c_2 < \cdots < c_5 \qquad \ldots(13)$$

where I(·) is the indicator function, equal to 1 when the condition holds and 0 otherwise.

No parameter of the stack, including the cut-points, was fitted on validation or test essays.

2.5 Personalized Feedback Generation

The framework generates feedback from five interpretable indicators: academic vocabulary ratio, grammar issue score, argument density, discourse marker density, and citation score. Reference thresholds were set at the median (50th percentile) of each feature's distribution in the training data only.

For academic vocabulary ratio, argument density, discourse marker density, and citation score, feedback is triggered when an essay's value falls below the median:

$$F_j^{(i)} = \mathrm{I}\big(l_{i,j} < \mathrm{Median}\big(L_j\big)\big) \qquad \ldots(14)$$

where L_j is the training distribution of feature j. Since a higher grammar issue score means more potential errors, the condition is reversed for grammar:

$$F_{\text{grammar}}^{(i)} = \mathrm{I}\big(l_{i,\text{grammar}} > \mathrm{Median}\big(L_{\text{grammar}}\big)\big) \qquad \ldots(15)$$

Each triggered condition maps to a predefined, actionable recommendation for the corresponding weakness. Comparing essays against training-population distributions rather than fixed thresholds identifies relative areas for improvement while keeping feedback tied to interpretable essay properties.

2.6 Evaluation Metric

Model performance is evaluated using Quadratic Weighted Kappa (QWK):

$$\text{QWK} = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \qquad w_{ij} = \frac{(i-j)^2}{(K-1)^2} \qquad \ldots(16)$$

where O is the observed confusion matrix between human and predicted scores, E is the expected matrix under independence (the outer product of the two score histograms, scaled to the same total), and K = 6 is the number of score levels.

QWK was computed on integer scores. Uncertainty was estimated with a 1,000-sample bootstrap over the test essays (percentile 95% interval).

3. Experiments and Results

3.1 Training and Inference Details

The DeBERTa and MiniLM encoders were frozen. The fusion network was trained with Adam (learning rate 1e-3, weight decay 1e-4, gradient clipping 1.0) for up to 20 epochs per fold with early stopping on validation RMSE. At inference, an essay passes through the same preprocessing and feature pipeline, the five fusion models and five ridge models are averaged, the level-1 models produce a continuous score, and the cut-points give the final integer score.

Table 1: Key hyperparameters of the proposed framework

Hyperparameter	**Value**
BiLSTM hidden size (per direction)	256
Regression head	128 hidden units, dropout 0.2
Word encoder	DeBERTa-v3-xsmall, max 512 tokens, 8-token blocks
Sentence/paragraph/chunk encoder	all-MiniLM-L6-v2
Projection size	128
Chunk size / stride (tokens)	56/25

Hyperparameter	Value
Fusion batch size	32
Fusion epochs / early-stopping patience	20/3 (per fold)
Optimizer	Adam, learning rate 1e-3, weight decay 1e-4
Auxiliary loss weight α	0.15
Out-of-fold folds (fusion, TF-IDF ridge, level-1)	5
TF-IDF	1-2 grams, min df 3, max 120,000 features; ridge alpha 1.0
LightGBM	learning rate 0.03, 15 leaves, min 40 samples per leaf
XGBoost	learning rate 0.03, depth 4, min child weight 20
Train/validation/test split	80:10:10, stratified, seed 42

3.2 Results

Table 2 reports QWK on the 1,731 held-out test essays. Published systems are evaluated on different datasets and protocols, so we compare only against variants of our own framework trained and evaluated on identical splits.

Table 2: Test QWK by model variant on the 1,731 held-out essays (cut-point scores are the reported result)

Variant	QWK (Rounded)	QWK (Cut-points)
Linguistic features only (GBM)	0.7266	0.7596
Linguistic + TF-IDF branch	0.7977	0.8137
Embedding branch only (BiLSTM fusion network)	0.8011	0.8096
Linguistic + embedding branch	0.7984	0.8132
Full model (linguistic + TF-IDF + embeddings)	0.8097	0.8206

The full model achieves the highest QWK (0.8206; 95% bootstrap CI 0.8030 to 0.8367), with validation QWK 0.8080 and test RMSE 0.5438. Its margin over the best single-branch variants (0.8137 and 0.8096) is small relative to the uncertainty of one test split of 1,731 essays, so we describe the gains as consistent but modest; no paired significance test was run.

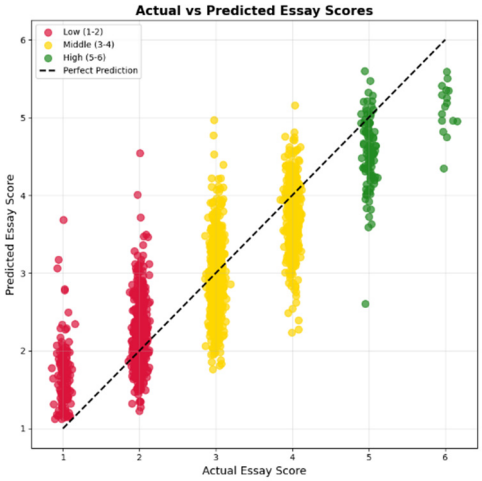
3.3 Ablation Study

Interpretable features alone reach a QWK of 0.7596 (0.7266 without cut-point optimization), so surface and lexical-quality indicators carry a substantial share of the signal but are not sufficient by themselves. The multi-granularity embedding branch alone reaches 0.8096, comparable to the linguistic + TF-IDF stack (0.8137), showing that frozen semantic representations capture much of what human scores depend on. Combining all three components gives 0.8206. The embedding branch adds 0.0069 QWK beyond the linguistic + TF-IDF stack (0.8206 vs. 0.8137; 0.8097 vs. 0.7977 without cut-points), which supports the use of multi-granularity semantic representations, although the gain is small compared with the confidence interval. Cut-point optimization improves every variant, by 0.009 to 0.033 QWK (for example, 0.7266 to 0.7596 for the linguistic features alone).

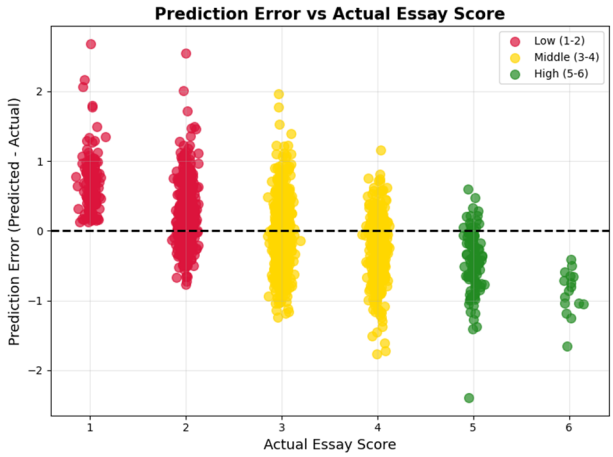
3.4 Prediction Analysis and Visualization

To look beyond aggregate QWK, two plots were generated on the held-out test set: predicted versus ground-truth scores (Figure 2a) and prediction error, defined as predicted minus ground truth, versus ground-truth scores (Figure 2b). Essays are grouped as Low (1-2), Middle (3-4), and High (5-6). The dashed diagonal in Figure 2a marks perfect agreement, and the dashed horizontal line in Figure 2b marks zero error. Predictions are compressed toward the middle of the scale: low-scoring essays tend to be over-predicted and high-scoring essays under-predicted. This is typical of

regression toward the mean and is strongest for score 6, which has few essays (156 of 17,307). The overall test RMSE is 0.5438.



(a) Predicted versus ground-truth essay scores



(b) Prediction errors across score categories

Figure 2: Test-set performance. (a) Predicted vs. ground-truth scores; dashed line is y = x. (b) Prediction error (predicted − actual) by score category

3.5 Deployment

The framework is deployed through a lightweight Gradio interface where a user pastes an essay and receives the predicted score, component scores (embedding branch, lexical branch, final), extracted linguistic features, and personalized feedback (Figures 3 and 4).

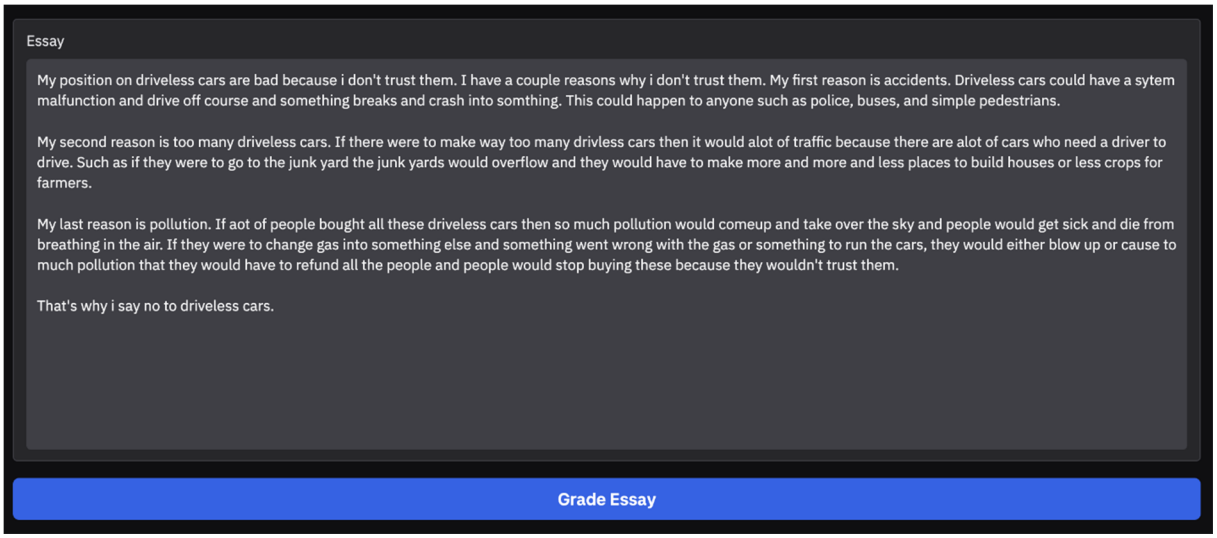
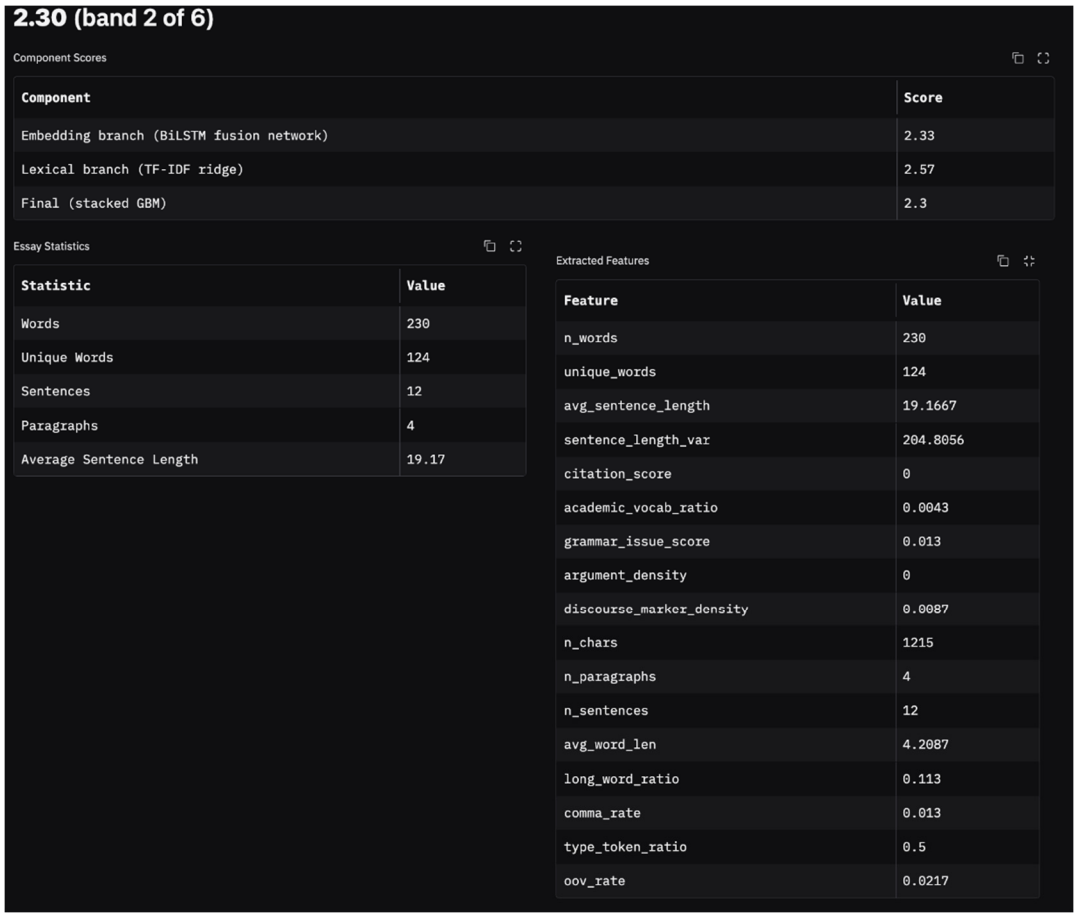
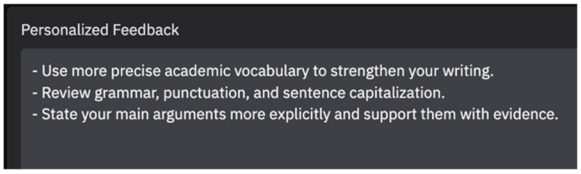


Figure 3: Essay input interface



(a) Score and statistics



(b) Personalized feedback

Figure 4: Output interface: (a) score and statistics; (b) personalized feedback

4. Discussion and Limitations

The framework combines interpretable linguistic features, a lexical signal, and hierarchical semantic representations, achieving a QWK of 0.8206 (95% CI 0.8030 to 0.8367) on 1,731 held-out essays. The ablations show that the embedding branch is the strongest single component (0.8096) and that stacking all components adds a modest further gain (+0.0069 over the linguistic + TF-IDF stack), which lies within the uncertainty of a single test split; repeated splits or a paired

bootstrap would be needed to confirm it. Several limitations remain. The model was trained and tested on a single public dataset of argumentative essays, so cross-prompt and out-of-domain generalization is untested. Interpretability is limited to the linguistic indicators and the feedback built from them; the embedding and TF-IDF branches, which drive most of the accuracy, are not directly interpretable. Several linguistic indicators (academic vocabulary, grammar issues, argument density, citation score) are heuristic approximations rather than validated measures. Length-related features can be gamed, and the model may reward length. Feedback thresholds are set at feature medians, so about half of all essays are flagged on each indicator regardless of quality, and the pedagogical value of the feedback has not been evaluated with learners or teachers. Results come from one data split, and QWK is dataset-dependent and should be read as contextual. Fairness across demographic groups and robustness to adversarial inputs need further study before high-stakes use.

5. Conclusion and Future Work

This work presents a multi-granularity, linguistically enriched AES framework that stacks out-of-fold BiLSTM-attention representations, a lexical branch, and interpretable linguistic features in gradient-boosted regressors with QWK-optimized cut-points. On the Learning Agency Lab Automated Essay Scoring 2.0 dataset it attains a QWK of 0.8206 and produces feature-based feedback for learners and educators.

Future work includes: (i) cross-prompt and cross-domain evaluation, including few-shot prompt adaptation; (ii) fairness, uncertainty, and calibration analyses with human-in-the-loop review; (iii) stronger explanation methods, such as counterfactual edits and stable feature attribution, tied to scoring rubrics; (iv) efficient deployment via distillation and lightweight encoders; and (v) multilingual extension and trait-level subscore prediction.

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